8.8 Improper Integrals

We have considered definite integrals of the form $I = \int_a^b f(x)dx$,

where the integrand f is ***continuous*** on the ***closed, finite*** interval $[a, b]$.

The integral I is necessarily a finite number. Such integrals are also called ***proper integrals***.

Definite integral: $I = \int_a^b f(x)dx$

- **Proper**
  f is continuous on the closed, finite interval $[a, b]$.

- **Improper**
  - Type I: $a \to -\infty$ or $b \to \infty$. *(Domain is infinite)*
  - Type II: $f(x) \to \pm\infty$ at some points on $[a, b]$. *(Range is infinite)*

Example:

$$\int_0^2 \frac{1}{1+x^2}\,dx$$

$$\int_1^\infty \frac{1}{x(1+x^2)}\,dx$$

$$\int_0^2 \frac{1}{1-x^2}\,dx$$

Remark:

If $f(x) \geq 0$ over an interval $[a, b]$, the area under the curve $y = f(x)$ is given by $\int_a^b f(x)dx$.

An improper integral (Type I or Type II) can also be interpreted as an area. The area is infinite if the improper integral diverges.

Type I Improper Integrals

Integrals with infinite limits of integration are **improper integrals of Type I.**

1. If $f(x)$ is continuous on $[a, \infty)$, then
$$\int_a^{\infty} f(x)\,dx = \lim_{b\to\infty} \int_a^b f(x)\,dx.$$
2. If $f(x)$ is continuous on $(-\infty, b]$, then
$$\int_{-\infty}^{b} f(x)\,dx = \lim_{a\to-\infty} \int_a^b f(x)\,dx.$$
3. If $f(x)$ is continuous on $(-\infty, \infty)$, then
$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{c} f(x)\,dx + \int_c^{\infty} f(x)\,dx,$$
where c is any real number.

In each case, if the limit is finite we say that the improper integral **converges** and that the limit is the **value** of the improper integral. If the limit fails to exist, the improper integral **diverges**.

Example 1

Is the area under the curve $y = (\ln x)/x^2$ from $x = 1$ to $x = \infty$ finite? If so, what is it?

Example 2

Evaluate $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$.

Example 3

For what values of p does the integral $\int_1^\infty \frac{1}{x^p} dx$ converge? When the integral does converge, what is its value?

Type II Improper Integrals: Integrals of functions that become infinite at a point within the interval of integration are **improper integrals of Type II.**

1. If $f(x)$ is continuous on $(a, b]$ and discontinuous at a, then
$$\int_a^b f(x)\,dx = \lim_{c \to a^+} \int_c^b f(x)\,dx.$$

2. If $f(x)$ is continuous on $[a, b)$ and discontinuous at b, then
$$\int_a^b f(x)\,dx = \lim_{c \to b^-} \int_a^c f(x)\,dx.$$

3. If $f(x)$ is discontinuous at c, where $a < c < b$, and continuous on $[a, c) \cup (c, b]$, then
$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx,$$
where c is any real number.

In each case, if the limit is finite we say that the improper integral **converges** and that the limit is the **value** of the improper integral. If the limit fails to exist, the improper integral **diverges**.

Example 4

Investigate the convergence of

$$\int_0^1 \frac{1}{1-x}\, dx.$$

Example 5

Evaluate $\int_0^3 \frac{dx}{(x-1)^{2/3}}$.

Example 6

$$\int_{2}^{\infty} \frac{x+3}{(x-1)(x^2+1)}\,dx = \ln 5 + \tan^{-1}(2) - \frac{\pi}{2} \approx 1.1458.$$

Caution: It is **<u>incorrect</u>** to write

- $\int_{-1}^{2} \frac{1}{x^2} dx = \left[-\frac{1}{x}\right]_{-1}^{2} = -\frac{3}{2}.$

- $\int_{0}^{4} \frac{1}{\sqrt{x}} dx = \left[2\sqrt{x}\right]_{0}^{4} = 4.$

Do not apply FTC when $f(x)$ is discontinuous on $[a, b]$.

The two examples above are **<u>Type II Improper Integrals.</u>**

★ When an integral with <u>finite</u> limits of integration are given, check whether it is proper or improper.

Example 7

Evaluate $\int_0^3 \frac{dx}{x-1}$.

Example 8

Does the integral $\int_1^{\infty} e^{-x^2}\,dx$ converge?

Tests for Convergence and Divergence

When an improper integral cannot be evaluated, we may still be able to determine whether the integral converges (or diverges) by comparing it with one integral whose convergence (divergence) properties are known.

Theorem: Direct Comparison Test

Let f and g be continuous on $[a, \infty)$ with $0 \leq f(x) \leq g(x)$ for all $x \geq a$. Then

1. $\int_a^\infty f(x)\,dx$ converges if $\int_a^\infty g(x)\,dx$ converges

2. $\int_a^\infty g(x)\,dx$ diverges if $\int_a^\infty f(x)\,dx$ diverges.

Example 9

- $\int_1^{\infty} \frac{\sin^2 x}{x^2} dx$ converges

- $\int_1^{\infty} \frac{1}{\sqrt{x^2 - 0.1}} dx$ diverges

Caution: <u>Do Not Use</u> the Direct Comparison Test to test $\int_a^\infty f(x)\,dx$ when

i. $f(x) \le g(x)$ and $\int_a^\infty g(x)\,dx$ diverges; or

ii. $g(x) \le f(x)$ and $\int_a^\infty g(x)\,dx$ converges.

For example, to determine if $\int_1^\infty \frac{1-e^{-x}}{x}dx$ converges, we cannot compare it with $\int_1^\infty \frac{1}{x}dx$ even though we know $\frac{1-e^{-x}}{x} \le \frac{1}{x}$. This is because $\int_1^\infty \frac{1}{x}dx$ diverges, and we cannot say anything about $\int_1^\infty \frac{1-e^{-x}}{x}dx$. (In particular, we cannot deduce from direct comparison test that it diverges.) We use limit comparison test instead.

Theorem: Limit Comparison Test

If the positive functions f and g are continuous on $[a, \infty)$ and if

$$\lim_{x \to \infty} \frac{f(x)}{g(x)} = L,\ 0 < L < \infty,$$

then $\int_a^\infty f(x)\,dx$ and $\int_a^\infty g(x)$ both converge or both diverge.

Example: Test $\int_1^\infty \frac{1-e^{-x}}{x}\,dx$ for convergence.

Example 10

Show that $\int_1^\infty \frac{dx}{1+x^2}$ converges by comparison with $\int_1^\infty \frac{1}{x^2}\,dx$. Find and compare the integral values.

Example 11

Show that $\int_1^{\infty} \frac{3}{e^x+5}\,dx$ converges.

Extra Examples:

1. $\int_0^{\ln 2} \frac{1}{x^2 e^{1/x}}\, dx$

2. $\int_{\pi}^{\infty} \frac{2+\cos x}{x}\, dx$

3. $\int_1^{\infty} \frac{1}{\sqrt{e^x - 1}}\, dx$